

AfyaFlow

Presentation

Hospital-Wide Outpatient Demand & Resource Allocation Prototype

A forecasting and simulation prototype for outpatient congestion, built on 26.5 months of Kijabe Hospital registration data.

Presented by:

The Queue Busters

The Team



Viola Rono

*Business Understanding
& Preprocessing*



Silvia Odhiambo

Data Cleaning



Peter Magondu

*Deployment &
Modeling*



Leon Oswago

*Data cleaning
& Modeling*



Catherine Kagwiria

Feature Engineering



Alisha Saburali

*Data Understanding
& Modeling*



The Problem

Outpatient demand is volatile — and Kenya's hospitals are already stretched

1:17,000

doctor-to-patient ratio in Kenya, vs. the WHO-recommended 1:1,000

1 in 4

patients wait over an hour just to be seen

70,000

more health workers needed per year than the 7,650 currently trained

At AIC Kijabe Hospital

Patient volumes swing sharply by day and department. Weekdays run far busier than weekends. The result: congestion, long waits, and strained staff — and in August 2026, Kenyatta National Hospital had to ask patients to seek care elsewhere.



Business Understanding

Business Objectives

- *To predict outpatient demand across departments*
- *To flag high-volume departments, days, and times*
- *To understand what drives demand changes*
- *To support smarter staff and resource allocation*
- *To give hospital-wide visibility for capacity planning*

Research Questions

- *Can we predict demand to reduce congestion?*
- *What time patterns drive that demand?*
- *Do recurring (returning) patients explain some of this congestion?*
- *Can simulation help test staffing decisions?*

Data Overview

Source: AIC Kijabe Hospital Outpatient Records System (Opd_data.csv). This is the sole dataset available for the project — there is no separate data yet on staffing levels or bed availability.

Our project followed the CRISP-DM methodology

306,306

Rows

6

Columns

26.5

Months of data

1

Hospital Dataset

June 1, 2024 → Aug 19, 2026 (latest month incomplete)

Six columns captured per visit

PatientNumber

Unique patient identifier, links repeat visits

QueuedTo

department/clinic the patient was queued to

Age

age at registration (mixed-unit text)

RegistrationDate

date & time registered — main temporal field

ConsultationDescription

type/purpose of the consultation

Gender

recorded as Male or Female



Data Understanding

Data quality at a glance

Everything arrives as text or dates — age is written like “38 Yr(s)”, department has 44 possible values, and visit type has 79. All data types are correct except Age, which needs numeric conversion.

80,928

Unique patients

40

Median patient age

3.8 x

Avg. visit per patient

Gender Split

Female 57%

Male 43%

39% visit only once · 7.7% visit 10+ times (chronic / recurring care)

7 of 44 departments drive ~70% of all visits. These departments are: General outpatient, MCH, Admission, Chronic Care, Speciality Clinic, Oncology and Renal

Data Cleaning

- Dropped rows missing both `QueuedTo` and `ConsultDescription` (~6,500)
- Removed 355 exact duplicate records
- Dropped ~10,300 timestamps stuck at 00:00:00 (default, not real arrivals)
- Dropped remaining missing Gender/Age (<0.2% of data)
- Standardized inconsistent text casing



Feature Engineering

From Raw Visits to Model -Ready features

Age column

split "44 Yr(s)"/"6 Mth(s)" text into number + unit, converted to years

Return visits

Flag for whether a patient has been seen before, then flagged True/False

Lag & rolling averages

shifted past visit_count backward (lag), averaged recent periods (rolling)

Service type

79 messy consult labels mapped into 7 clean service groups.

Calendar features

extracted hour, day_of_week, is_weekend, month from RegistrationDate

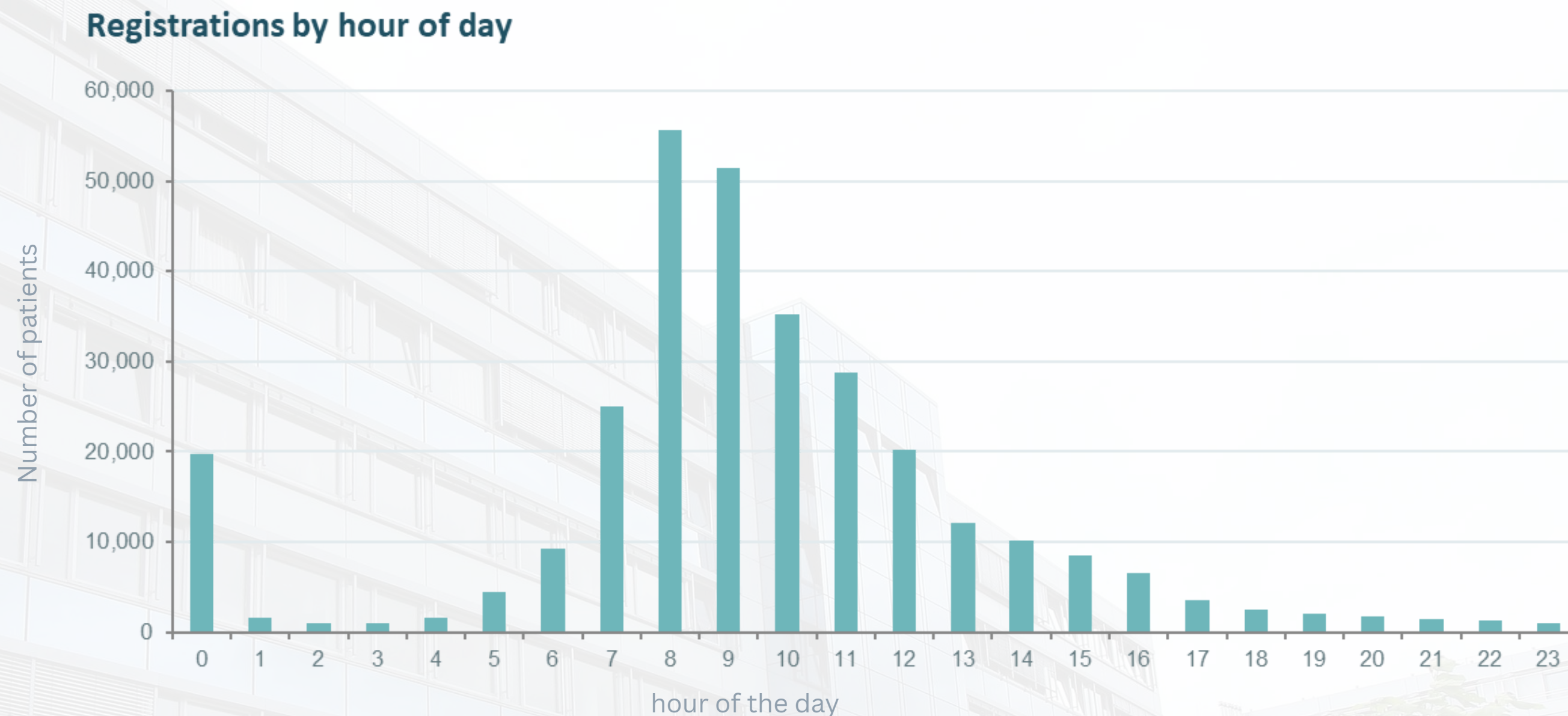
Hourly demand

grouped visits by QueuedTo + hour, counted into visit_count



Exploratory data analysis

Univariate Analysis



Busiest window: 07:00–14:00, peaking at 08:00–09:00 (55,567 registrations)

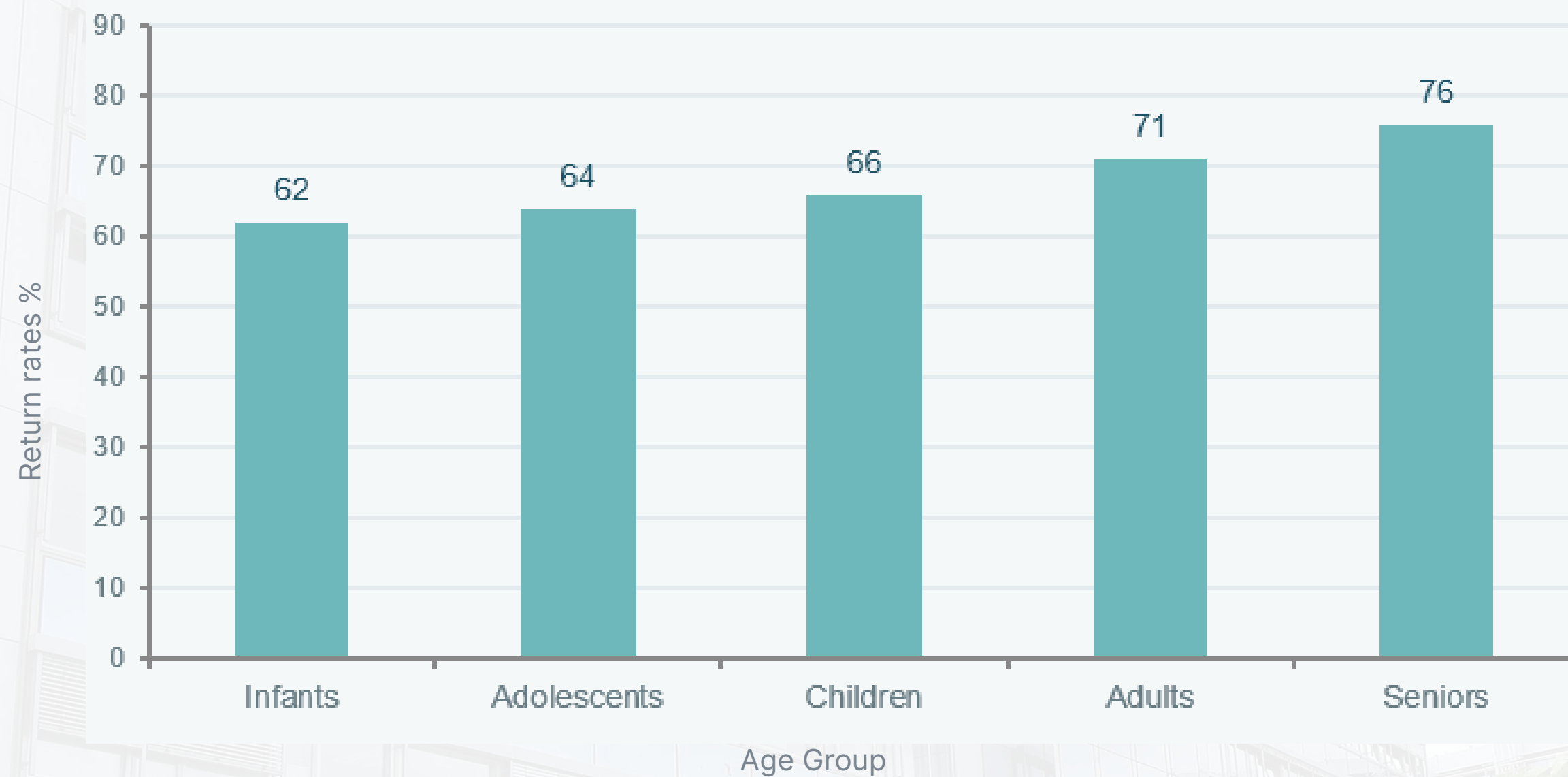
Majority of visits fall under General Outpatient Care, Admission, and CONSULT — consistent with General OPD being the single busiest department (85,959 visits).



Explratory data anlysis

Bivariate analysis

Return Rates by Age Group

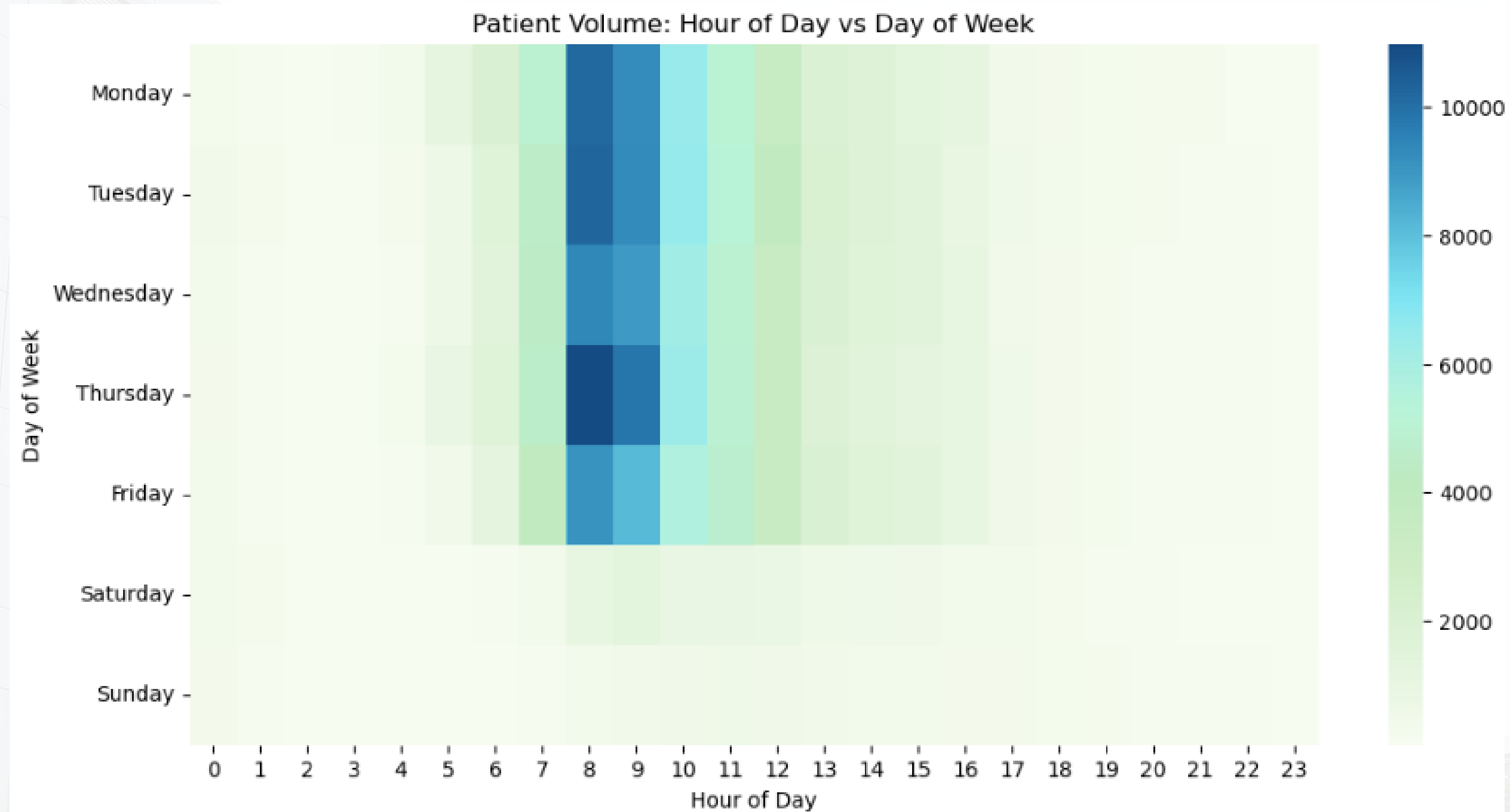


Return rates rise steadily with age – Seniors return most and infants the least



Exploratory data analysis

Multivariate analysis



The 80:00-10:00 peak holds across every weekday. High on Monday, Tuesday and Thursday and fades off almost completely on the weekends



Modeling



Seasonal Naive

Our starting point:
Predicts next week using last week's actual numbers.

Why: a simple, honest baseline. Every other model had to beat this to be worth using.

LightGBM

A machine-learning model that studies years of hospital visits to learn busy vs. quiet patterns.

Why: strong at picking up complex day/department patterns a simple rule would miss.

Prophet

A forecasting model built specifically for tracking trends and patterns over time.

Why: purpose-built for time-based forecasting, so it catches broader trends LightGBM can miss.

Ensemble

For each department, we use whichever of the two models works best for it.

Why: no single model wins everywhere. Combining them plays to each one's strengths.

Each model is tested against the one before it. A new model only stays if it actually predicts better.



Model Architecture	Modeling Logic & Approach	MAE (Patients / Wk)	RMSE (Patients / Wk)	R2	Operational Impact
Seasonal Naive (baseline)	Predicts current week using exact prior week actual demand	11.05	15.32	0.724	Reference baseline carried from prior week
Prohphet	Additive time-series tracking macro trends & weekly seasonality	10.54	14.58	0.751	+4.6% improvement over baseline
LightGBM	Gradient boosted decision trees using lag, rolling & date features	10.00	13.85	0.785	+9.5% improvement overall (18–26% drop in hourly error)
Hybrid Departmental Ensemble	Optimal model selection per department based on historical error	8.96	12.94	0.842	+18.9% top performance (Best model)



Conclusion

- *Outpatient demand is genuinely predictable from time-of-day, day-of-week, recent demand, and department identity.*
- *Analysis of patient records reveals that outpatient congestion at AIC Kijabe Hospital is heavily concentrated, with major morning arrival surges reaching 35-59 visits/hour*
- *LightGBM beats the seasonal-naïve baseline where it matters most operationally — hourly (18–26% error reduction) and daily (19–22%) forecasts.*
- *Patient age and consult type are meaningfully associated with return-visit behavior; gender is not — supporting the alternative hypothesis.*

Recommendations

- *Increase staffing on weekday mornings, especially Monday and the 08:00–09:00 peak,*
- *Prioritize resources to the 7 departments (General OPD, Admission, MCH, Chronic Care, Speciality Clinic, Oncology, Renal) that drive ~70% of visits.*
- *Use the hourly/daily forecasts for staffing*
- *Renal Dialysis and Program Patients return at 99%+ rates. Preschedule slots for these cohorts in order to free walk-in capacity.*
- *Use the LightGBM hourly and daily models — not the monthly one — to drive short-term staffing and room allocation; monthly numbers should stay a high-level planning reference only.*
- *Implement digital self-service check-in tools at registration to reduce check-in time*



Thank You

